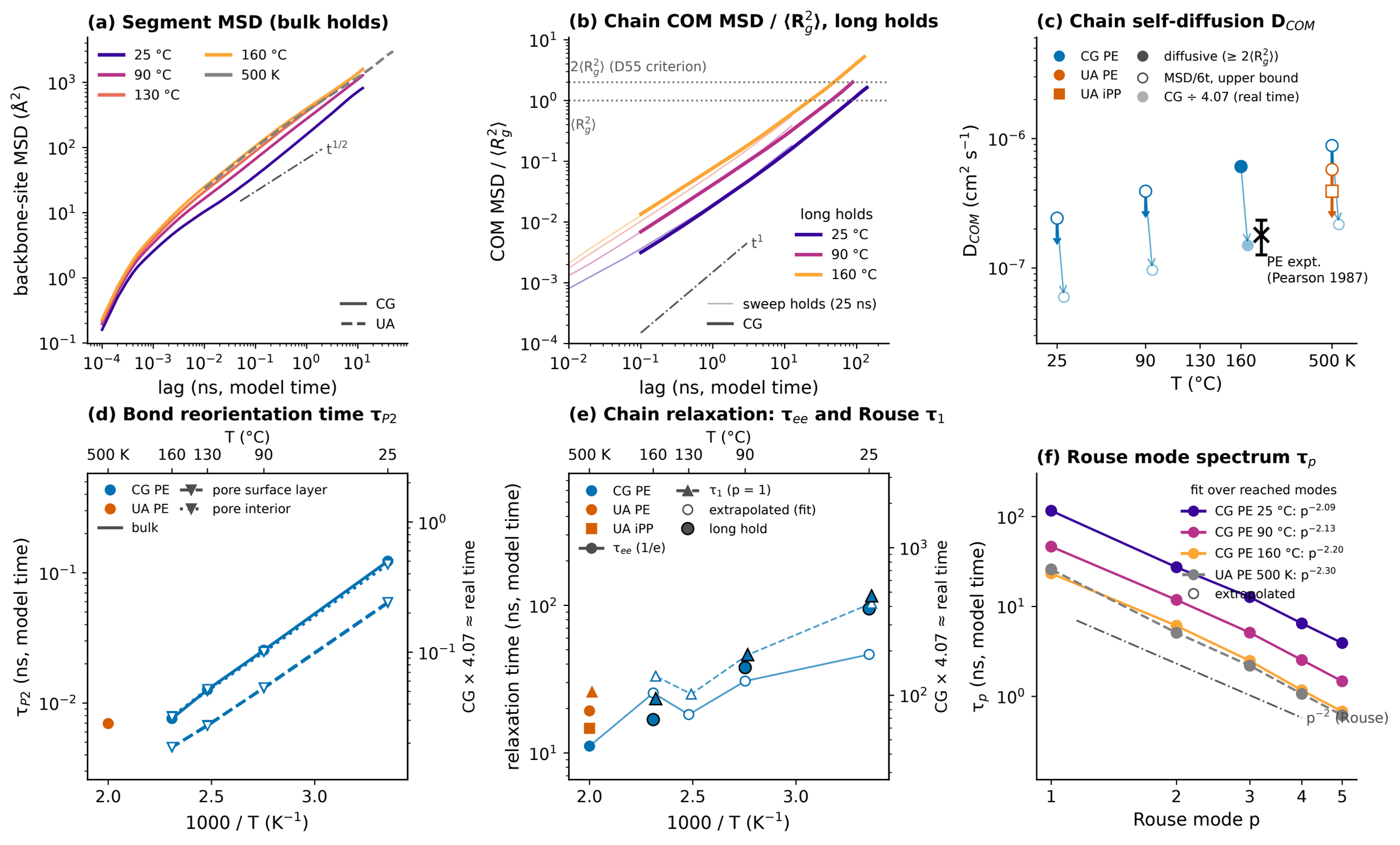

(a) Segment MSD (bulk holds)
25 °C
90 °C
130 °C
160 °C
500 K
CG
UA
t^1/2
backbone-site MSD (Å²)
lag (ns, model time)
(b) Chain COM MSD / ⟨R_g²⟩, long holds
2⟨R_g²⟩ (D55 criterion)
⟨R_g²⟩
long holds
25 °C
90 °C
160 °C
sweep holds (25 ns)
CG
t^1
COM MSD / ⟨R_g²⟩
lag (ns, model time)
(c) Chain self-diffusion D_COM
CG PE
UA PE
UA iPP
diffusive (≥ 2⟨R_g²⟩)
MSD/6t, upper bound
CG ÷ 4.07 (real time)
PE expt.
(Pearson 1987)
D_COM (cm² s⁻¹)
T (°C)
(d) Bond reorientation time τ_P2
T (°C)
CG PE
UA PE
bulk
pore surface layer
pore interior
τ_P2 (ns, model time)
CG × 4.07 ≈ real time
1000 / T (K⁻¹)
(e) Chain relaxation: τ_ee and Rouse τ_1
T (°C)
CG PE
UA PE
UA iPP
τ_ee (1/e)
τ_1 (p = 1)
extrapolated (fit)
long hold
relaxation time (ns, model time)
CG × 4.07 ≈ real time
1000 / T (K⁻¹)
(f) Rouse mode spectrum τ_p
fit over reached modes
CG PE 25 °C: p^−2.09
CG PE 90 °C: p^−2.13
CG PE 160 °C: p^−2.20
UA PE 500 K: p^−2.30
extrapolated
p^−2 (Rouse)
τ_p (ns, model time)
Rouse mode p